

From Genes to Cell Types: A Network-Based Investigation of ASD-ADHD Comorbidity

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1

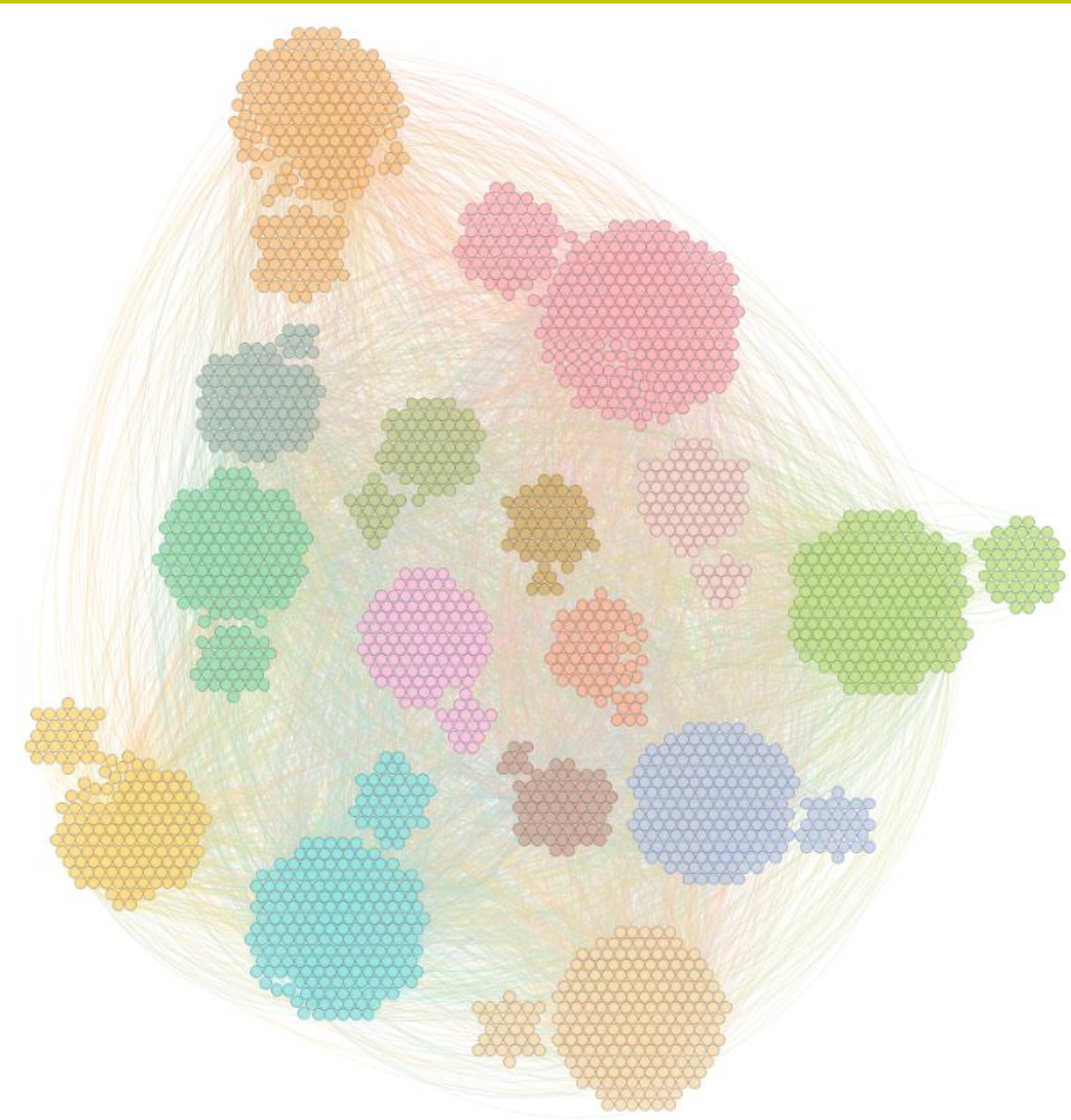
COMMUNITY CLUSTERING

2

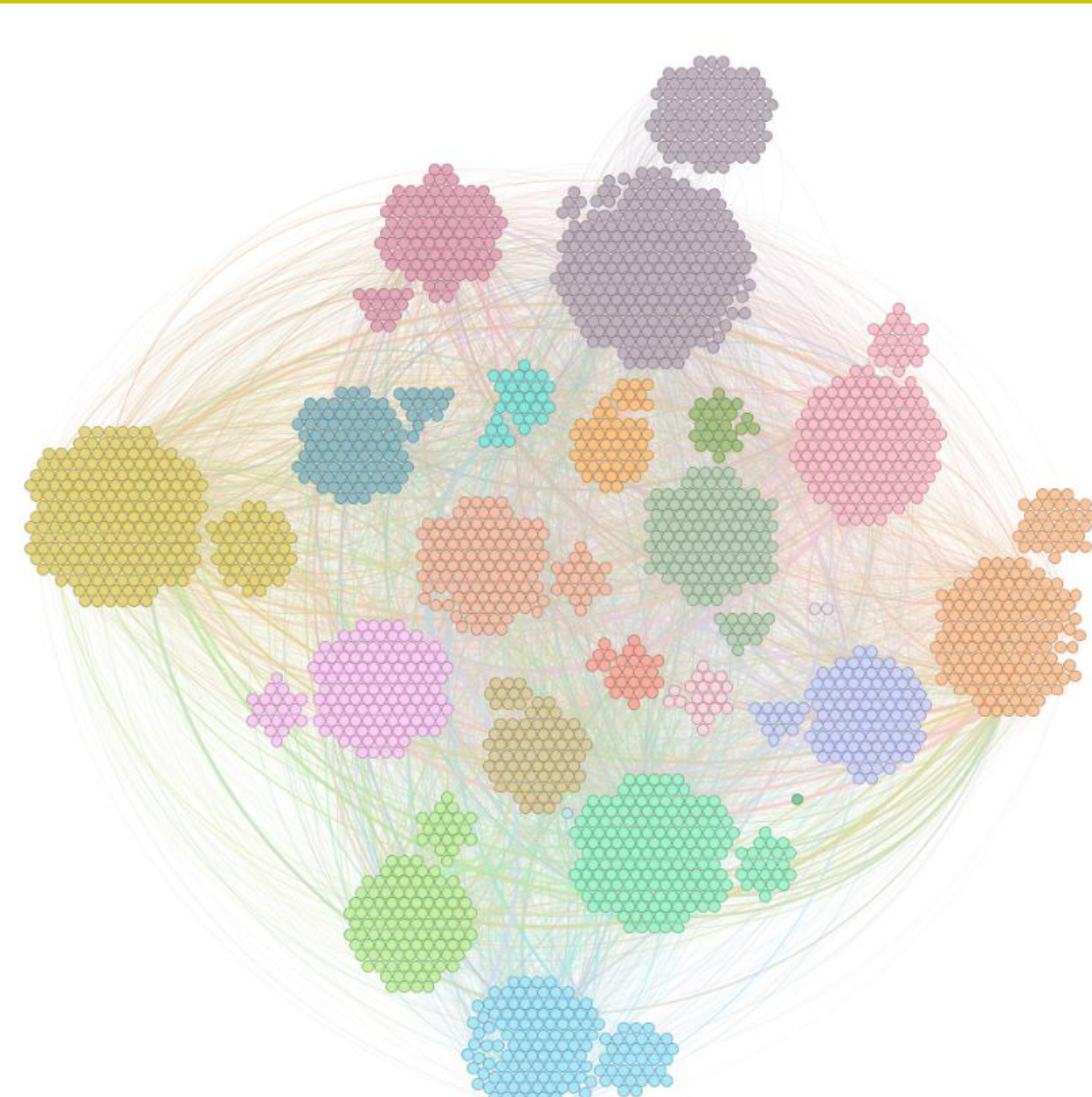
DISEASE OVERREPRESENTATION

3

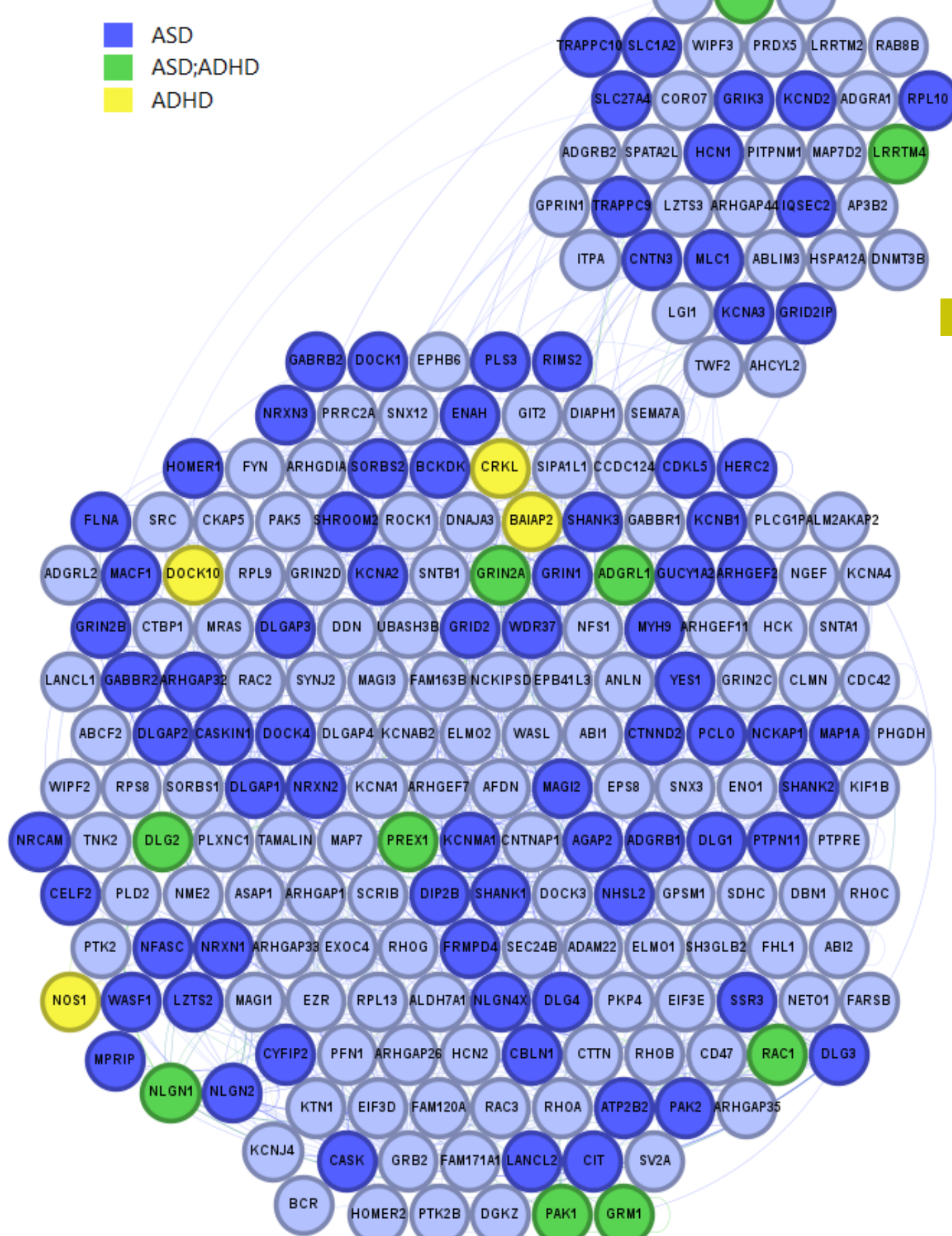
FUNCTIONAL ENRICHMENT



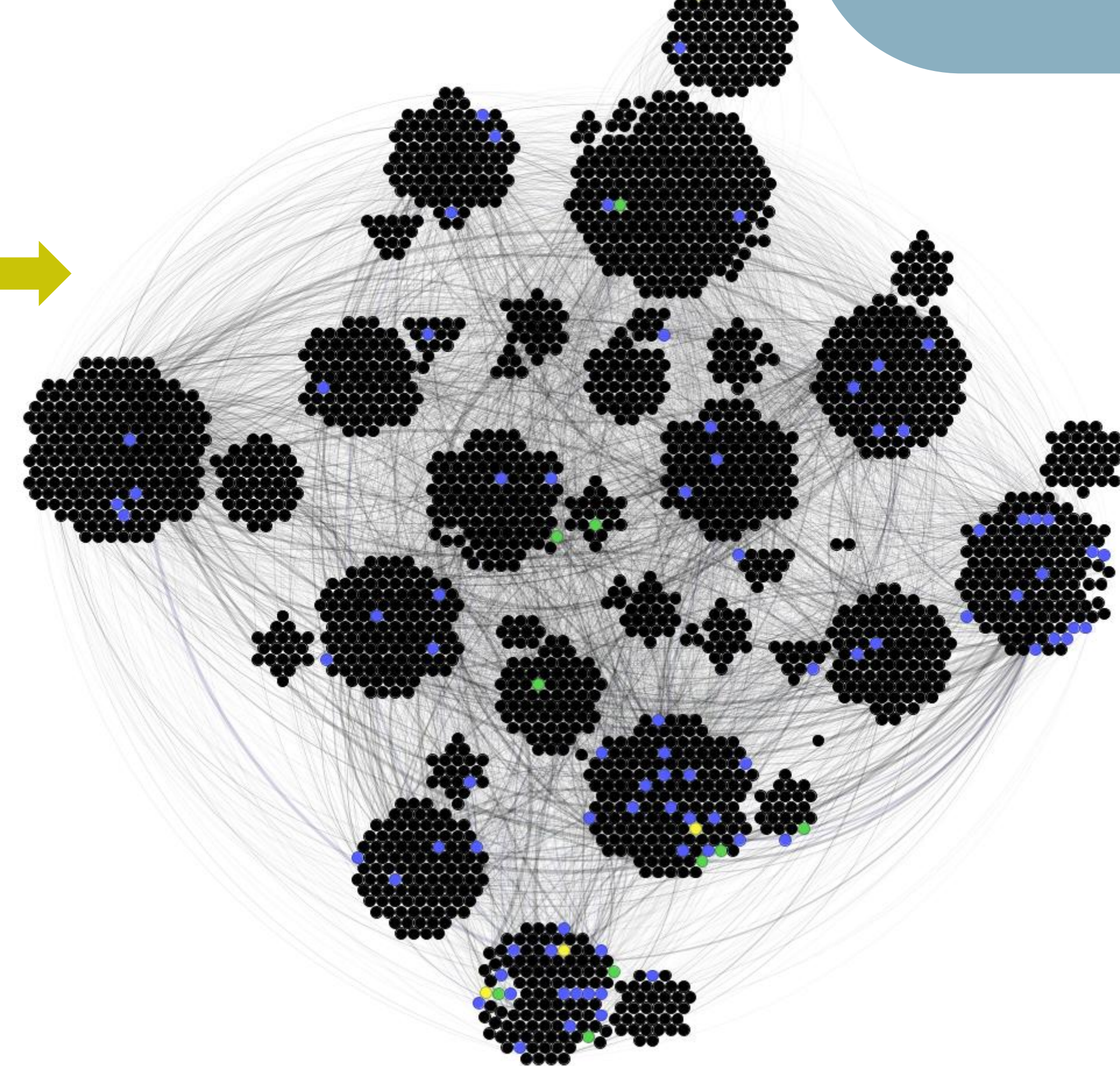
UNWEIGHTED NETWORK
2297 nodes
15 clusters of which:
3 enriched for ASD
1 enriched for ADHD



MICROGLIA NETWORK
2288 nodes
22 clusters of which:
2 enriched for ASD



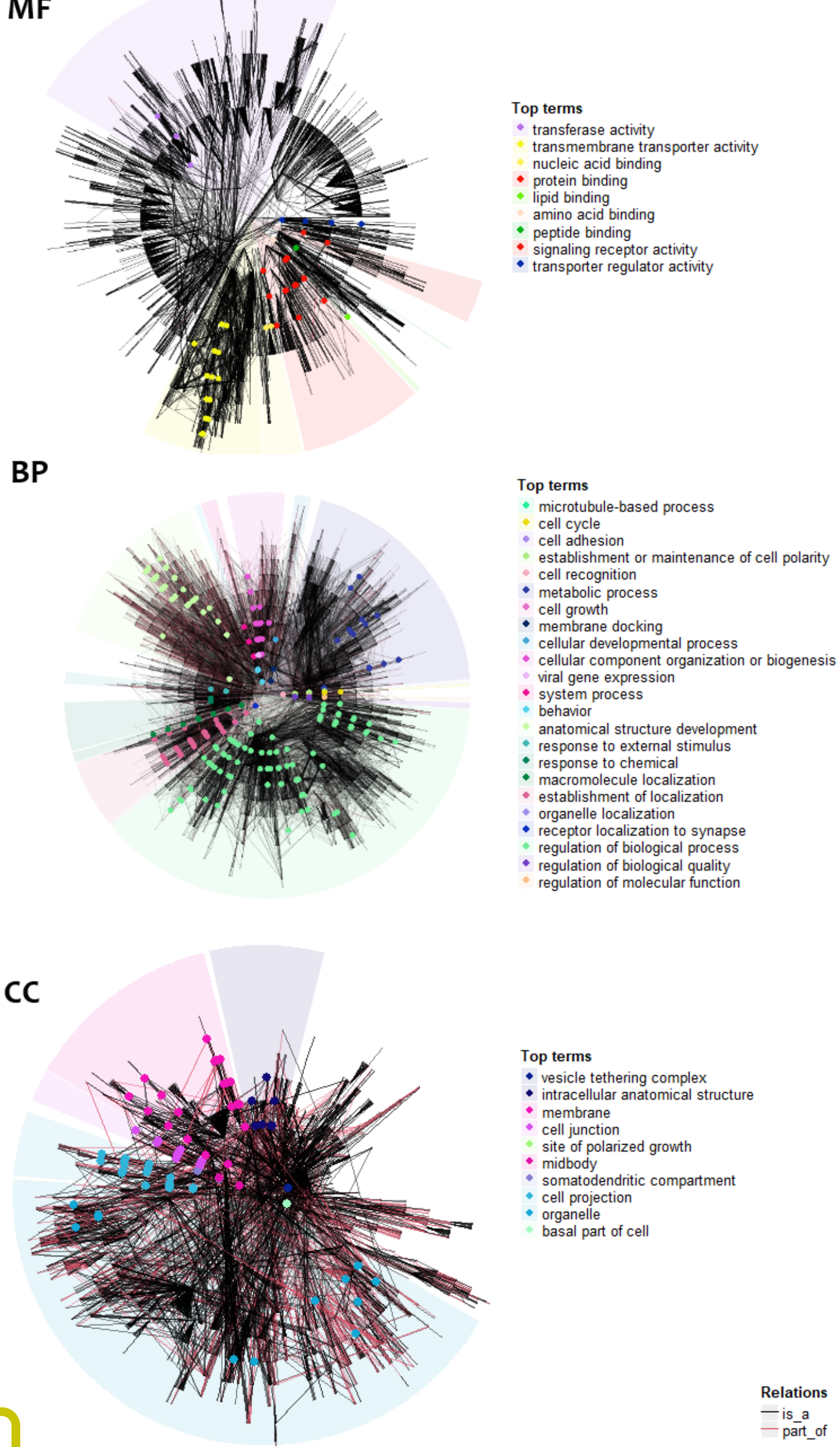
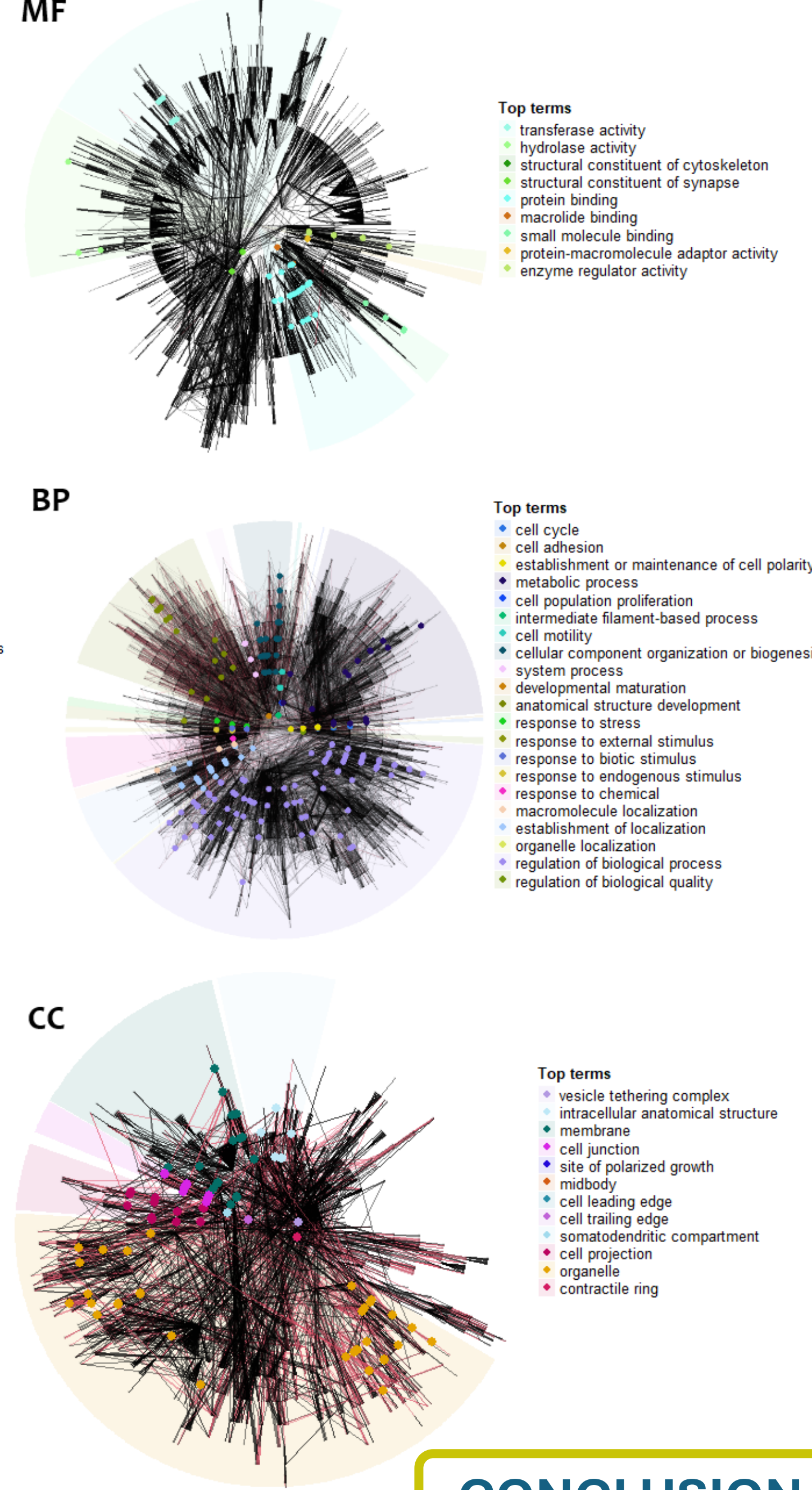
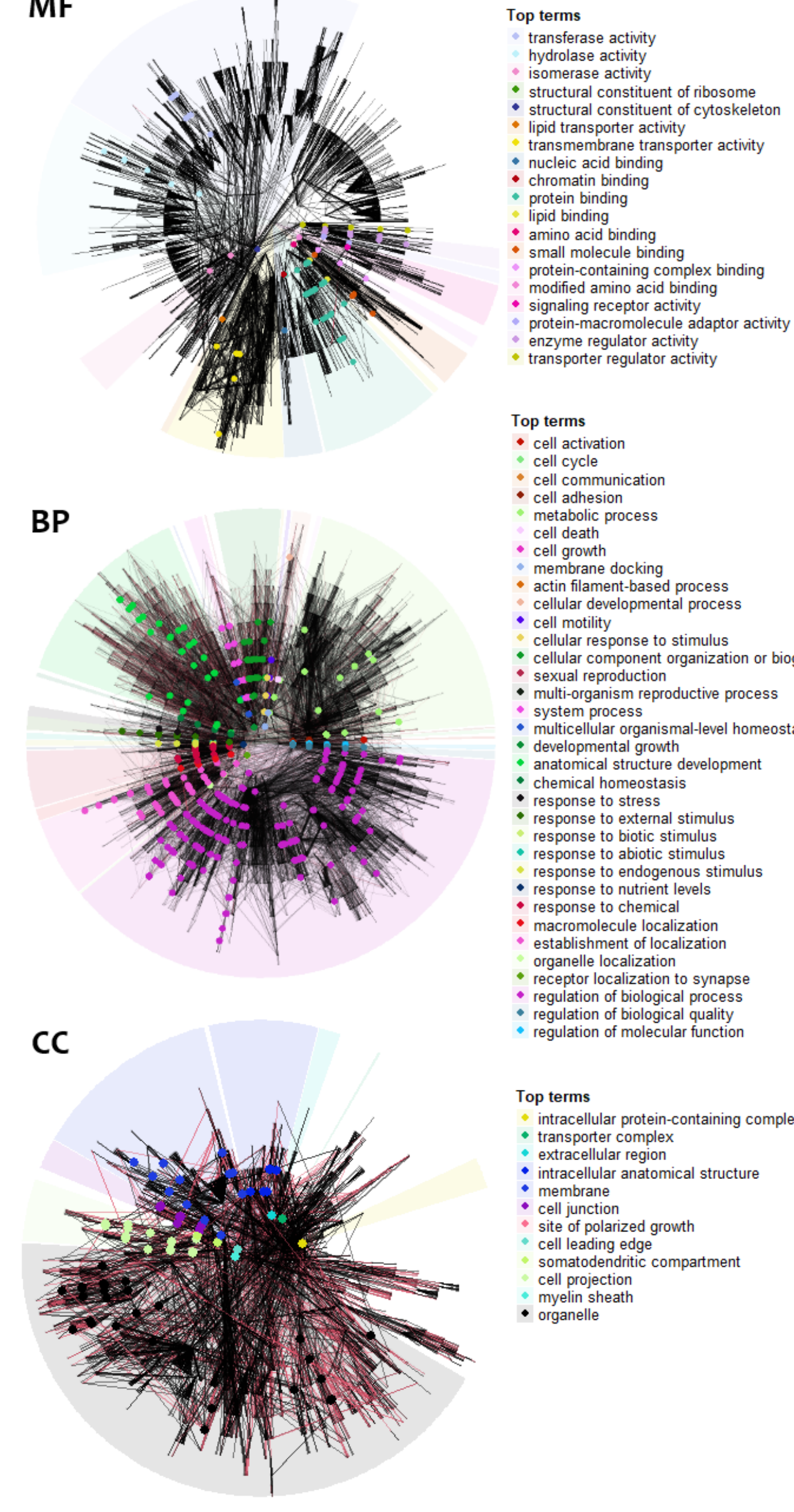
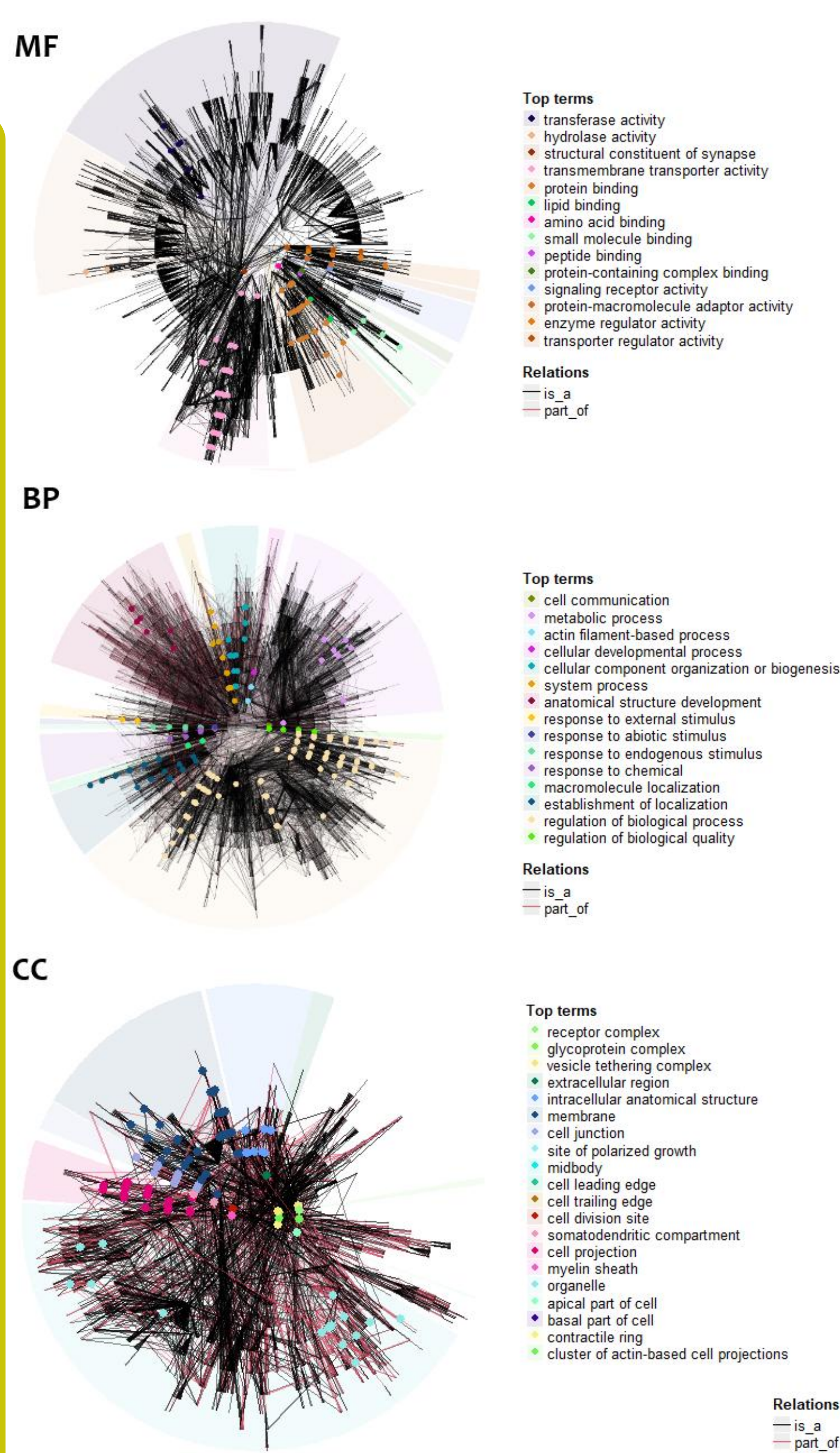
UNWEIGHTED BASELINE CLUSTER 7



MICROGLIA CLUSTER 4

MICROGLIA CLUSTER 8

MICROGLIA CLUSTER 10



CONCLUSION

This analysis of the microglia cell type has demonstrated that the integration of cell-type specific expression data into the PPI networks can significantly alter community structure and disease overrepresentation. Moreover, it can showcase cell-type specific functional modules or pathways involved in the conditions despite lack of one-to-one cluster matching. While our initial results highlight intriguing patterns in non-neuronal cell types, which warrant further exploration, analysis of neuronal cell types is prioritised to uncover shared biological pathways between ASD and ADHD. This analysis enables us to shed light on the complex relationships between genetic factors, cellular mechanisms, and phenotypic outcomes in comorbid ASD and ADHD.

Gene Ontology (GO) Enrichment Analysis

- Molecular Function (MF):** Notably, peptide binding is unique to baseline cluster 7 and microglia cluster 10. In ASD, this may relate to theories of neuroinflammation and findings of altered synaptic pruning through phagocytosis and degradation.
- Biological Processes (BP):** While overlapping terms are observed, there are no unique terms shared between baseline and microglia cluster 10.
- Cellular Components (CC):** The term 'basal part of the cell' is the only uniquely shared term between baseline cluster 7 and microglia cluster 10. This could potentially affect microglia motility and function.

References:

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- Bastian M., Heymann S., Jacomy M. (2009). Gephi: an open source software for exploring and manipulating networks. International AAAI Conference on Weblogs and Social Media.
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QR code links to project GitHub and contact details

